

Bridge Chloride Exposure Predictor: a web-based tool that to forecast the future chloride exposure rates for bridges

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Summary

The Bridge Chloride Exposure Predictor (BCEP) is a web-based tool designed to forecast future chloride exposure rates for bridges based on a calculation model developed by Xu and Yang (Xu, 2024; Xu et al., 2024), with an assumption that the climate and traffic data is constant within a specific GRID_SIZE for a given JURISDICTION. This model integrates traffic and climate data to predict chloride exposure, specifically focusing on damage from deicing salts while excluding other factors such as mechanical wear or accidents. Currently, this model is validated for Ontario, Canada. A [demonstration website](#) is available for an accessible overview. The tool can be used to investigate chloride exposure for new jurisdictions by providing new traffic and climate data. The inputs, outputs, data requirements and theoretical models for BCEP are summarized in the [Software Requirements Specification \(SRS\)](#).

Statement of need

Most highway bridges are constructed with reinforced concrete decks, which are susceptible to chloride-induced corrosion due to adverse weather conditions and traffic patterns. Various factors contribute to this corrosion, including the quality of construction materials and maintenance practices, with deicing salts being a significant contributor. The primary deicing salt used is sodium chloride (rock salt). When rock salt melts the snow and comes into contact with water, it can undergo a chemical reaction, releasing chloride ions. These ions can penetrate the concrete and cause corrosion in the reinforcing steel, thereby compromising the bridge's structural integrity and load-bearing capacity. Chloride is considered the main source of corrosion damage to reinforced concrete. Chloride ions are transported from the road to the exterior surface of bridge substructures through vehicle spray and splash mechanisms. There is a tight connection between chloride exposure, weather conditions and traffic flow. Specifically, the amount of deicing salt applied on the road surface greatly depends on the amount of snowfall, and the amount of water and dissolved chloride ions that end up on nearby objects depends on the traffic patterns.

While existing research explores the relationship between chloride and corrosion, there remains a lack of accessible research tool to visualize chloride exposure rates effectively. Accurate prediction of chloride exposure rates is essential for effective bridge management, construction planning, and research. Government agencies can use this data to prioritize maintenance and allocate budgets efficiently, focusing on bridges with higher corrosion risks. Bridge engineers can use the information to determine minimum structural requirements and ensure that bridges can endure their intended lifespan, particularly during the precise design phase. Researchers, especially those studying the impacts of climate change on infrastructure, benefit from predictive models to understand and mitigate corrosion damage. This project, conducted in collaboration

41 with Dr. Cancan Yang from McMaster University, highlights the importance of such tools in
42 addressing the challenges posed by chloride-induced corrosion.

43 State of the field

44 Proprietary tools exist for predicting and designing the service life of different concrete bridge
45 designs, such as [Life-365](#), [STADIUM](#) and [COMSOL](#) models. However, these tools require an
46 estimate of the expected chloride concentration in the environment around the bridge. As far
47 as we know BCEP is the first tool to determine and visualize this value for a given location via
48 local traffic and climate data.

49 Software design

50 BCEP consists of Python code and JavaScript with Vue. The Python code is for converting
51 the raw climate and traffic data into a database covering the jurisdiction of interest. The
52 JavaScript and Vue are used for the web application, which provides a graphical view for
53 input and output. Javascript and Vue were chosen because they integrate well with map
54 data, allowing for visualization and mouse click input that is geographically restricted. Vue
55 simplifies UI development by introducing a declarative programming model. The [installation](#)
56 [instructions](#) are for a Python virtual environment, so installation is isolated to not interfere
57 with the developer's usual build environment. The Python and JavaScript combination make
58 the software portable between Windows, Mao OS, and Linux.

59 The design of the software is based on the principle of information hiding ([David L. Parnas, 1972](#)). That is, the design is decomposed around likely changes, which become the secrets of
60 the modules. Likely changes include the algorithm for calculating chloride exposure and the
61 method for graphing the data. The software is divided into 17 modules, including modules
62 for input, control, constants, chloride on pier calculations and plotting. The details of the
63 decomposition, and the mapping between the conceptual design and the source code, can
64 be found in the [Module guide](#). The module guide, which includes the uses relation between
65 modules, is based on Parnas's ideas for documenting the software architecture ([D. L. Parnas et al., 1984](#); [David L. Parnas & Weiss, October 2001](#)). The interface provided by each module
66 is documented in the [Module Interface Specification \(MIS\)](#). The MIS follows the approach
67 presented by Hoffman and Strooper ([Hoffman & Strooper, 1995](#)) and later adapted for scientific
68 computing software ([ElSheikh et al., 2004](#); [Smith & Yu, 2009](#)).

71 BCEP has been verified following the [Verification and Validation Plan](#), results of which are
72 show in the [Verification and Validation Report](#).

73 Research impact statement

74 BCEP was developed to assist with Xu's PhD research "[Lifetime Serviceability and Safety of Highway Bridges Under Climate Change](#)" ([Xu, 2024](#)). Xu's dissertation addresses the growing
75 challenge of ensuring the serviceability and safety of concrete highway bridges amidst climate
76 change. Using BCEP future researchers will be able to continue to investigate the effects of
77 climate change on bridges safety. Xu's research focused on Canada, but the tool was designed
78 to be usable in other jurisdictions by [updating the traffic and climate data](#).

80 Although not the main purpose of BCEP, there is a secondary research impact. BCEP was
81 used as a case study for applying software engineering principles (like information hiding) and
82 methodologies (like modular decomposition) to scientific computing software. This work is
83 documented in the report "[A Collaborative Framework Toward Minimizing Pain Points in Research Software Development](#)" ([Liu, 2024](#)). The report describes a practical framework to
84 empower small teams of domain experts and developers to collaboratively build sustainable
85

research software. The framework addresses common pain points of evolving requirements and researchers' limited technical familiarity. Central to the approach is structured requirements elicitation, where developers guide domain experts through targeted questions about theories, typical uses cases, computational problem scale and possible tests. The answers directly inform modular design, verification, and documentation.

Features and usage

A quick start guide is provided in the [README.md](#) of the repository.

As outlined in [Figure 1](#), there are two sources of input:

- User Input: Information is entered through the website interface on the front end.
- Developer Input: Updates (if any) to the traffic table, climate table, and salt application rate need to be made by editing the code directly.

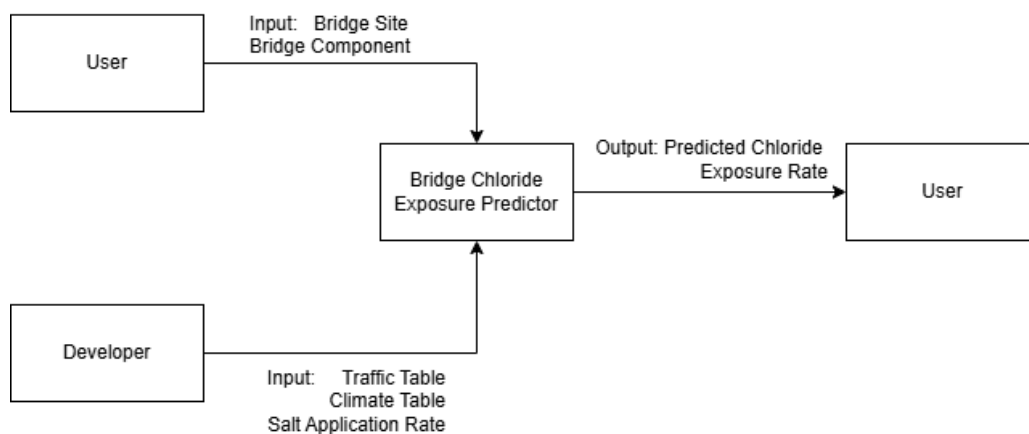


Figure 1: Input and Output of BCEP

The following sections provide guidance on modifying and customizing the website to suit other jurisdictions.

Customization

To adapt the code for jurisdictions beyond Ontario, follow these steps to ensure compatibility and proper results:

1. Adapting Mechanisms

- The physical mechanisms remain the same, but differences in deicing salt application policies should be considered when customizing for a new jurisdiction.

2. Data Collection and Preparation

- Collect climate data and traffic data specific to the target jurisdiction.
- Organize the data into a single Excel file.
- Ensure Consistency:
 - **Grid Size Matching:** The grid sizes for climate and traffic data must align.

- 113 – **Row Alignment:** Each row in the sheets should represent the same location. For
114 instance, row 2 across all sheets must correspond to the same longitude and latitude.

115 2.1 Climate Data Requirements

- 116 ■ **Data Types:**
117 – `htotal`: Total snowfall during winter
118 – `t1`: Number of days with snowfall
119 – `t2`: Number of days with snow melting
120 ■ **Data Format:**
121 Climate data should be structured as shown below, with years as columns:

Year	2006	2007	2008	...
	206.2145241	220.8943212	121.5795313	...
	211.174729	221.0476551	137.5310097	...

122 2.2 Traffic Data Requirements

- 123 ■ **Data Types:**
124 – `AADT/Lane`: Annual average daily traffic per lane
125 – `AADTT/Lane`: Annual average daily truck traffic per lane
126 ■ **Data Format:**
127 Traffic data should include coordinates, city, and traffic metrics per lane:

Longitude	Latitude	City	AADT/Lane	AADTT/Lane
274.712677	49.45780945	Algoma	559	103
275.9333496	47.5466156	Algoma	559	103
276.9502258	50.10984039	Cochrane	603	132

128 3. Sheet Naming Conventions

129 To ensure the tool functions correctly, the Excel file must follow specific naming conventions
130 for each sheet:

- 131 ■ **Climate Data Sheets:**
132 – Each type of climate data (`htotal`, `t1`, `t2`) should be placed in a separate sheet,
133 as shown in the data format above.
134 – The sheet names must exactly match the data type, e.g., a sheet for total snowfall
135 should be named `htotal`.
136 ■ **Traffic Data Sheet:**
137 – All traffic data (`AADT`, `AADTT`) should be included in a single sheet, as shown in the
138 data format above.
139 – The sheet must be named `traffic`.

140 AI Usage Disclosure

141 AI was not used for coding or documentation. However, AI was used for brainstorming content
142 for the [CONTRIBUTING.md](#) guide, the [CodeOfConduct.md](#) and the continuous deployment
143 of the documentation via a [GitHub Action](#). In all cases the tool used was ChatGPT based on
144 GPT-5.5. The human authors have reviewed, verified and edited all AI suggested text.

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